

McKenzie McCain

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EDUCATION

Texas A&M University – College Station, TX

May 2027

B.S. in Bionics and Robotics, Interdisciplinary Track

- **Minors:** Biomedical Engineering (biomechanics) **Cumulative GPA:** 3.96/4.00
- **Organizations:** Biomedical Engineering Society (BMES), Engineers without Borders (EWB), Aggie Blossoms

Relevant Courses: Geometric Modeling, Engineering & Newtonian Mechanics, Calculus I-III, Differential Equations

Licenses & Certifications: Additive Manufacturing: Optimizing 3D Prints, Learning 3D Printing

PROJECTS

EMG-Controlled Prosthetic Hand | *T.U.R.T.L.E. Robotics OLSN Project* | College Station, TX

May 2025 – Present

- Collaborated with the research and development of an electromyographic-controlled prosthetic hand for a local boy
- Utilized SolidWorks design tables to parameterize a scan of the client's arm, manufacture a socket, and create a mechanical hand.

Control and Robotics (CTRLRobot) Lab | *Undergraduate Researcher* | Advisor: Dr. Minghui Zheng

December 2025 – Present

- Conducted research into a 3D printable biomimetic robotic gripper, implemented calibration tests with Dynamixel servos and integrated force-sensing resistors for feedback control using Embedded C++ and Arduino.

Hatchling Universal Robot Controller (H.U.R.C.) | *T.U.R.T.L.E. Robotics Development Program*

May 2025 – Present

- Designed and built seven ESP32-based controllers for competition robots, using ESP-NOW P2P Wi-Fi pairing (up to 20 peers).
- Developed the TurtleReceiver software library in C++ and integrated an OLED debug display for simple and optimized functions (triggers/bumpers/joysticks/XYAB) to streamline controller integration for newer team members.

Electrophysiology Medical Device | *MedXplore Competition* | College Station, TX

January 2026

- Built an exoskeleton knee brace to provide controlled electrical stimulation and reduce the patient return-to-lifestyle period for lower-extremity muscle atrophy; placing 2nd in the division. Utilized embedded C++ to control a 3-planet gearbox with 20kg servos.

ACTIVITIES

T.U.R.T.L.E. Robotics | College Station, TX

Hatchling Director

April 2025 – Present

- Oversaw 100+ college students from various organizations and taught topics such as soldering, embedded programming, Solidworks/computer-aided design, and mentored individual competition teams

Hatchling Member

December 2024 – April 2025

- Designed a robot from scratch, utilizing Solidworks to 3-D model and print robotic components—placed 2nd against 21 competitors
- Worked with electronics (ESP32, L298n, DC-motors, and SG90 Micro Servos) to configure a custom electronic/mechanical system; utilized Arduino to program PWM motor control and mechanism functions to stack 2.5in blocks

Aggie Blossoms | College Station, TX

September 2024 – Present

- Collaborated in planning events that promote empowerment and support among women students
- Served my community through social activities and community events

Biomedical Engineering Society | College Station, TX

January 2025 – Present

- Networked with companies in the BMEN industry during weekly meetings; Improved professional skills and participated in mentorship program events

Freshman Leaders in New Technology (FLiNT) | College Station, TX

October 2023 – April 2024

- Organized campus events promoting student engagement and community involvement
- Collaborated with peers on projects, fostering leadership growth and personal development

EXPERIENCE

Rex Academy | *Programming Summer Camp Instructor (Contract)* | Southlake, TX

June 2025 – July 2025

- Developed lesson plans in the Rex Academy TLP; taught Scratch Jr. (K-2) and 2D Game Programming (3-5) with Flowlab
- Managed student accounts/devices and classroom setup; created a quick-start login guide and resolved bug/connectivity issues.
- Supervised and coached students to produce multiple playable mini-games and scene animations by week's end

Cinepolis Cinema | *Cast Member* | Euless, TX

March 2022 – July 2022

- Provided high-quality customer service to over 200 patrons daily, performed opening/closing duties, and managed inventory supplies

Colleyville Baseball Association | *Cashier and Concessions Supplier* | Colleyville, TX

August 2019 – May 2021

- Provided excellent customer service, assisting 250+ patrons and accurately balancing the register, processing up to \$1000 per shift
- Maintained a welcoming environment and managed inventory/restocking of supplies, handling up to \$600 deliveries

SKILLS

Tools & Platforms: SolidWorks, Autodesk (Fusion 360, Inventor), Visual Studio Code, Git/Github, Microsoft Office (Word, Excel, PowerPoint), Google Workspace (Docs, Sheets, Slides)

Technical: CAD/Design (Finite Element Analysis, 3-D Printing, Additive Manufacturing), **Embedded Systems** (Circuit Design, Microcontroller Programming (Embedded C)), **Robotics & Controls** (Controls Systems), **Programming** (Python, C/C++, Java)

Engineering & Analytical: Product Design/Development, Technical Report Writing, Mechanical System Design, Problem Solving